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DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING
B.E. ECE - REGULATIONS 2024 (R2024) - CHOICE BASED CREDIT SYSTEM
TOTAL PROGRAM CREDITS: 164

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VISION

To be a leading hub in Electronics and Communication Engineering, driving innovation, interdisciplinary collaboration, and socially impactful solutions through cutting-edge Education, Research and Entrepreneurship.

MISSION

1. To advance knowledge and practice in Electronics and Communication Engineering through hands-on, industry-relevant education that prepares students for innovation, entrepreneurship and multidisciplinary collaboration.
2. To foster a culture of research and development that addresses real-world challenges and creates transformative solutions at the intersection of Electronics, Communication and Computing.
3. To empower individuals and communities by leveraging technology for positive social impact through inclusive education, collaborative outreach and interdisciplinary teamwork.

PROGRAM EDUCATIONAL OBJECTIVES (PEOs)

PEO1. Graduates will have a thorough grounding in the fundamental sciences, facilitating future academic pursuits in ECE.
PEO2. Graduates will demonstrate expertise in ECE, empowering them to excel in industry applications, advanced studies, and research.
PEO3. Graduates will have a spirit of inquiry and learning, staying current with industry trends and technological breakthroughs.
PEO4. Graduates will critically assess literature, identify knowledge gaps, and develop novel, ethics-guided research approaches.
PEO5. Graduates will integrate professional ethics with social awareness, addressing engineering challenges holistically.

PROGRAM OUTCOMES (POs)

PO1. Engineering knowledge
PO2. Problem analysis
PO3. Design/development of solutions
PO4. Conduct investigations of complex problems
PO5. Modern tool usage

PO6. The engineer and society
PO7. Environment and sustainability
PO8. Ethics
PO9. Individual and team work
PO10. Communication
PO11. Project management and finance
PO12. Life-long learning

PROGRAM SPECIFIC OUTCOMES (PSOs)
PSO1. Apply electronic, mathematical, and engineering principles to design, develop, and analyse sophisticated electronic systems.
PSO2. Design, simulate, and optimize communication systems.
PSO3. Leverage the latest advancements in electronics and communication to design and develop groundbreaking solutions.

CATEGORY LEGEND

HSMC - Humanities, Social Sciences and Management
BSC - Basic Science Courses
ESC - Engineering Science Courses
PCC - Professional Core Courses
PEC - Professional Elective Courses
OEC - Open Elective Courses
EEC - Employability Enhancement Courses
MC - Mandatory Courses (non-credit)

Column format: L-T-P (Lecture-Tutorial-Practical periods/week)

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SEMESTER I (Total Credits: 22)
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S.No	Code	Course Title	Cat	L-T-P	Credits
1	U24EN101	Professional English - I	HSMC	3-0-0	3
2	U24MA101	Engineering Mathematics - I	BSC	3-1-0	4
3	U24PH101	Engineering Physics	BSC	3-0-0	3
4	U24CH101	Engineering Chemistry	BSC	3-0-0	3
5	U24CS101	Problem Solving and Python Programming	ESC	3-0-0	3
6	U24TM101	Heritage of Tamils	HSMC	1-0-0	1
7	U24CS1L1	Problem Solving and Python Programming Lab	ESC	0-0-4	2
8	U24BS1L1	Physics and Chemistry Laboratory	BSC	0-0-4	2
9	U24EN1L1	English Laboratory	HSMC	0-0-2	1
10	U24MC101	Induction Program (Mandatory, 3 weeks)	MC	-	-

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SEMESTER II (Total Credits: 26)

S.No	Code	Course Title	Cat	L-T-P	Credits
1	U24EN201	Professional English - II	HSMC	2-0-0	2
2	U24MA201	Engineering Mathematics - II	BSC	3-1-0	4
3	U24PH201	Applied Physics	BSC	3-0-0	3
4	U24ME201	Engineering Graphics	ESC	2-2-0	4
5	U24TM201	Tamils and Technology	HSMC	1-0-0	1
6	U24EE206	Basic Electrical and Instrumentation Eng.	ESC	3-0-0	3
7	U24EC201	Circuit Analysis	PCC	3-1-0	4
8	U24ME2L1	Engineering Practices Laboratory	ESC	0-0-4	2
9	U24EN2L1	Communication Skills Laboratory	HSMC	0-0-4	2
10	U24EC2L1	Circuit Analysis Laboratory	PCC	0-0-3	2
11	U24MC201	Universal Human Values (Mandatory)	MC	2-0-0	0

SEMESTER III (Total Credits: 23)

S.No	Code	Course Title	Cat	L-T-P	Credits
1	U24MA301	Engineering Mathematics - III	BSC	3-1-0	4
2	U24CS304	C Programming and Data Structures	ESC	3-0-0	3
3	U24EC301	Signals and Systems	PCC	3-0-0	3
4	U24EC302	Electronic Devices and Circuits	PCC	3-0-0	3
5	U24EC303	Control Systems	PCC	3-0-0	3
6	U24EC304	Digital Systems Design	PCC	3-0-2	4
7	U24EC3L1	Electronic Devices and Circuits Laboratory	PCC	0-0-3	1.5
8	U24CS3L4	C Programming and Data Structures Laboratory	ESC	0-0-3	1.5
9	U24MC301	Indian Traditional Knowledge (Mandatory)	MC	2-0-0	0
10	U24TP301	Career Enhancement Course - I	EEC	2-2-0	0

SEMESTER IV (Total Credits: 20)

S.No	Code	Course Title	Cat	L-T-P	Credits
1	U24EC401	Electromagnetic Fields	PCC	3-0-0	3
2	U24EC402	Linear Integrated Circuits	PCC	3-0-0	3
3	U24EC403	Communication Systems	PCC	3-0-0	3
4	U24BS401	Environmental Sciences and Sustainability	BSC	3-0-0	3
5	U24EC404	Digital Signal Processing	PCC	3-0-2	4
6	U24EC4L1	Communication Systems Laboratory	PCC	0-0-3	1.5
7	U24EC4L2	Linear Integrated Circuits Laboratory	PCC	0-0-3	1.5
8	U24MC401	Indian Constitution (Mandatory)	MC	2-0-0	0
9	U24TP401	Career Enhancement Course - II	EEC	2-2-0	0
10	U24TP402	Microsoft Essentials	EEC	0-0-2	1

SEMESTER V (Total Credits: 22)

S.No	Code	Course Title	Cat	L-T-P	Credits
1	U24EC501	VLSI and Chip Design	PCC	3-0-0	3
2	U24EC502	Transmission Lines and RF Systems	PCC	3-0-0	3
3	U24EC503	Microcontroller and Interfacing	PCC	3-0-0	3
4	U24ECPXX	Professional Elective - I	PEC	3-0-0	3
5	U24YYOEX	Open Elective - I	OEC	3-0-0	3
6	U24EC504	Wireless Communication and Network Security	PCC	3-0-2	4
7	U24EC5L1	VLSI Laboratory	PCC	0-0-4	2
8	U24MC501	Personality Development for Professionals	MC	2-0-0	0
9	U24TP5XX	Technical Training Course - I	EEC	0-0-2	1

SEMESTER VI (Total Credits: 19)

S.No	Code	Course Title	Cat	L-T-P	Credits
1	U24EC601	Optical Communication and Network	PCC	3-0-0	3
2	U24ECPXX	Professional Elective - II	PEC	3-0-0	3
3	U24YYOEX	Open Elective - II	OEC	3-0-0	3
4	U24EC602	Embedded System and IOT Design	PCC	3-0-2	4
5	U24CS610	Basics of Artificial Intelligence and ML	ESC	3-0-2	4
6	U24MC601	Creativity, Innovation and Entrepreneurship	MC	2-0-0	0
7	U24EM601	Mini Project	EEC	0-0-2	1
8	U24TP6XX	Technical Training Course - II	EEC	0-0-2	1

SEMESTER VII (Total Credits: 19)

S.No	Code	Course Title	Cat	L-T-P	Credits
1	U24BA701	Professional Ethics and Managerial Skills	HSMC	2-0-0	2
2	U24EC701	Remote Sensing	PCC	3-0-0	3
3	U24ECPXX	Professional Elective - III	PEC	3-0-0	3
4	U24ECPXX	Professional Elective - IV	PEC	3-0-0	3
5	U24YYOEX	Open Elective - III	OEC	3-0-0	3
6	U24YYOEX	Open Elective - IV	OEC	3-0-0	3
7	U24EM701	Internship (4 weeks)	EEC	-	2

SEMESTER VIII (Total Credits: 12)

S.No	Code	Course Title	Cat	L-T-P	Credits
1	U24EM801	Project Work	EEC	0-0-24	12

PROFESSIONAL ELECTIVES (PEC) - FOUR VERTICALS

(Students choose 4 electives - PE I to PE IV - across Sem V, VI & VII)

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--- VERTICAL I: SIGNAL PROCESSING ---

1	U24ECP11	Advanced Digital Signal Processing	3-0-0	3
2	U24ECP12	Image Processing	3-0-0	3
3	U24ECP13	Speech Processing	3-0-0	3
4	U24ECP14	Software Defined Radio	3-0-0	3
5	U24ECP15	DSP Architecture and Programming	3-0-0	3
6	U24ECP16	Computer Vision	3-0-0	3

--- VERTICAL II: SEMICONDUCTOR CHIP DESIGN AND TESTING ---

7	U24ECP21	Wide Band Gap Devices	3-0-0	3
8	U24ECP22	Validation and Testing Technology	3-0-0	3
9	U24ECP23	Low Power IC Design	3-0-0	3
10	U24ECP24	VLSI Testing and Design For Testability	3-0-0	3
11	U24ECP25	Analog IC Design	3-0-0	3
12	U24ECP26	Mixed Signal IC Design Testing	3-0-0	3

--- VERTICAL III: RF TECHNOLOGIES ---

13	U24ECP31	RF Transceivers	3-0-0	3
14	U24ECP32	Signal Integrity	3-0-0	3
15	U24ECP33	Antenna Design	3-0-0	3
16	U24ECP34	MICs and RF System Design	3-0-0	3
17	U24ECP35	EMI/EMC Pre-Compliance Testing	3-0-0	3
18	U24ECP36	RFID System Design and Testing	3-0-0	3

--- VERTICAL IV: SPACE TECHNOLOGIES ---

19	U24ECP41	Radar Technologies	3-0-0	3
20	U24ECP42	Avionics Systems	3-0-0	3
21	U24ECP43	Positioning and Navigation Systems	3-0-0	3
22	U24ECP44	Satellite Communication	3-0-0	3
23	U24ECP45	Rocketry and Space Mechanics	3-0-0	3
24	U24ECP46	Remote Sensing	3-0-0	3

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OPEN ELECTIVE COURSES (OEC) - OFFERED ACROSS DEPARTMENTS
(ECE students select Open Elective I-IV from any department below)

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--- ELECTRICAL AND ELECTRONICS ENGINEERING ---

1	U24EEOE1	Non-Conventional Energy Sources	3-0-0	3
2	U24EEOE2	Special Purpose Machines	3-0-0	3
3	U24EEOE3	Battery Management	3-0-0	3
4	U24EEOE4	Computer Control of Industrial Processes	3-0-0	3
5	U24EEOE5	Electric Vehicles	3-0-0	3
6	U24EEOE6	Energy Technology	3-0-0	3

--- ELECTRONICS AND COMMUNICATION ENGINEERING ---

7	U24ECOE1	Microprocessor and Microcontroller	3-0-0	3
8	U24ECOE2	Wireless Sensors and Networks	3-0-0	3
9	U24ECOE3	Sensors and Actuators	3-0-0	3
10	U24ECOE4	Mobile Communications	3-0-0	3
11	U24ECOE5	Embedded Systems	3-0-0	3
12	U24ECOE6	Nano Technology	3-0-0	3

--- MECHANICAL ENGINEERING ---

13	U24MEOE1	Industrial Robotics	3-0-0	3
14	U24MEOE2	Additive Manufacturing	3-0-0	3
15	U24MEOE3	Safety Engineering	3-0-0	3
16	U24MEOE4	Equipment for Pollution Control	3-0-0	3
17	U24MEOE5	Energy Conservation in Industries	3-0-0	3
18	U24MEOE6	Composite Materials and Mechanics	3-0-0	3

--- COMPUTER SCIENCE AND ENGINEERING ---

19	U24CSOE1	Computer Network Fundamentals	3-0-0	3
20	U24CSOE2	Basics of Cloud Computing	3-0-0	3

--- INFORMATION TECHNOLOGY ---

21	U24ITOE1	Web Design Concepts using HTML/CSS	3-0-0	3
22	U24ITOE2	Basics of PHP and MySQL	3-0-0	3

--- ARTIFICIAL INTELLIGENCE AND DATA SCIENCE ---

23	U24DSOE1	Fundamentals of Neural Networks	3-0-0	3
24	U24DSOE2	Data Science Fundamentals	3-0-0	3

--- CYBER SECURITY ---

25	U24CYOE1	Fundamentals of Cyber Forensic	3-0-0	3
26	U24CYOE2	Fundamentals of Ethical Hacking	3-0-0	3

--- ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING ---

27	U24MLOE1	Basics of Java	3-0-0	3
28	U24MLOE2	Basics of Linux Operating Systems	3-0-0	3

--- AGRICULTURAL ENGINEERING ---

29	U24AEOE1	Climate Change and Adaptation	3-0-0	3
30	U24AEOE2	Urban Agriculture	3-0-0	3
31	U24AEOE3	Automation in Agriculture	3-0-0	3
32	U24AEOE4	IT in Agriculture	3-0-0	3
33	U24AEOE5	Agriculture Entrepreneurship Development	3-0-0	3
34	U24AEOE6	Traditional Indian Foods	3-0-0	3

--- BIOMEDICAL ENGINEERING ---

35 U24BMOE1 Lifestyle Diseases 3-0-0 3
36 U24BMOE2 Pharmaceutical Nanotechnology 3-0-0 3
37 U24BMOE3 Biotechnology in Healthcare 3-0-0 3
38 U24BMOE4 Therapeutic Equipment 3-0-0 3
39 U24BMOE5 Medical Innovation and Entrepreneurship 3-0-0 3
40 U24BMOE6 Medical Instrumentation 3-0-0 3

--- MASTER OF BUSINESS ADMINISTRATION ---

41 U24BAOE1 Entrepreneurship Development 3-0-0 3
42 U24BAOE2 Basics of Accounting 3-0-0 3
43 U24BAOE3 Disaster Management 3-0-0 3
44 U24BAOE4 Introduction to Stock Market 3-0-0 3
45 U24BAOE5 Total Quality Management 3-0-0 3
46 U24BAOE6 Human Resource Management 3-0-0 3

TECHNICAL TRAINING COURSES (EEC)

--- TECHNICAL TRAINING COURSE - I (Semester V) ---

1 U24TP501 Advanced Python Programming 0-0-2 1
2 U24TP502 C, C++, Java Advanced Programming 0-0-2 1
3 U24TP503 Advanced Embedded Systems with IoT 0-0-2 1
4 U24TP504 CAD Foundations and Advanced Modeling 0-0-2 1

--- TECHNICAL TRAINING COURSE - II (Semester VI) ---

5 U24TP601 Advanced Data Analytics 0-0-2 1
6 U24TP602 Advanced Web Development Stack 0-0-2 1
7 U24TP603 Industrial Applications of Embedded Systems with Robotics 0-0-2 1
8 U24TP604 Industrial CAD and IoT 0-0-2 1

SUMMARY OF CREDIT DISTRIBUTION

Subject Area	Sem-I	Sem-II	Sem-III	Sem-IV	Sem-V	Sem-VI	Sem-VII	Sem-VIII	Total	%
HSMC	5	5	-	-	-	-	2	-	12	7
BSC	12	7	4	3	-	-	-	-	26	16
ESC	5	9	4.5	-	-	4	-	-	22.5	14
PCC	-	6	14.5	16	15	7	3	-	61.5	38
PEC	-	-	-	-	3	3	6	-	12	7
OEC	-	-	-	-	3	3	6	-	12	7
EEC	-	-	-	1	1	2	2	12	18	11
MC	(non-credit mandatory courses across semesters)								-	-
TOTAL	22	26	23	20	22	19	19	12	164	100

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Source: ECE R2024 Final Curriculum and Syllabus document (Anna University affiliated, Choice Based Credit System), Department of Electronics and Communication Engineering.